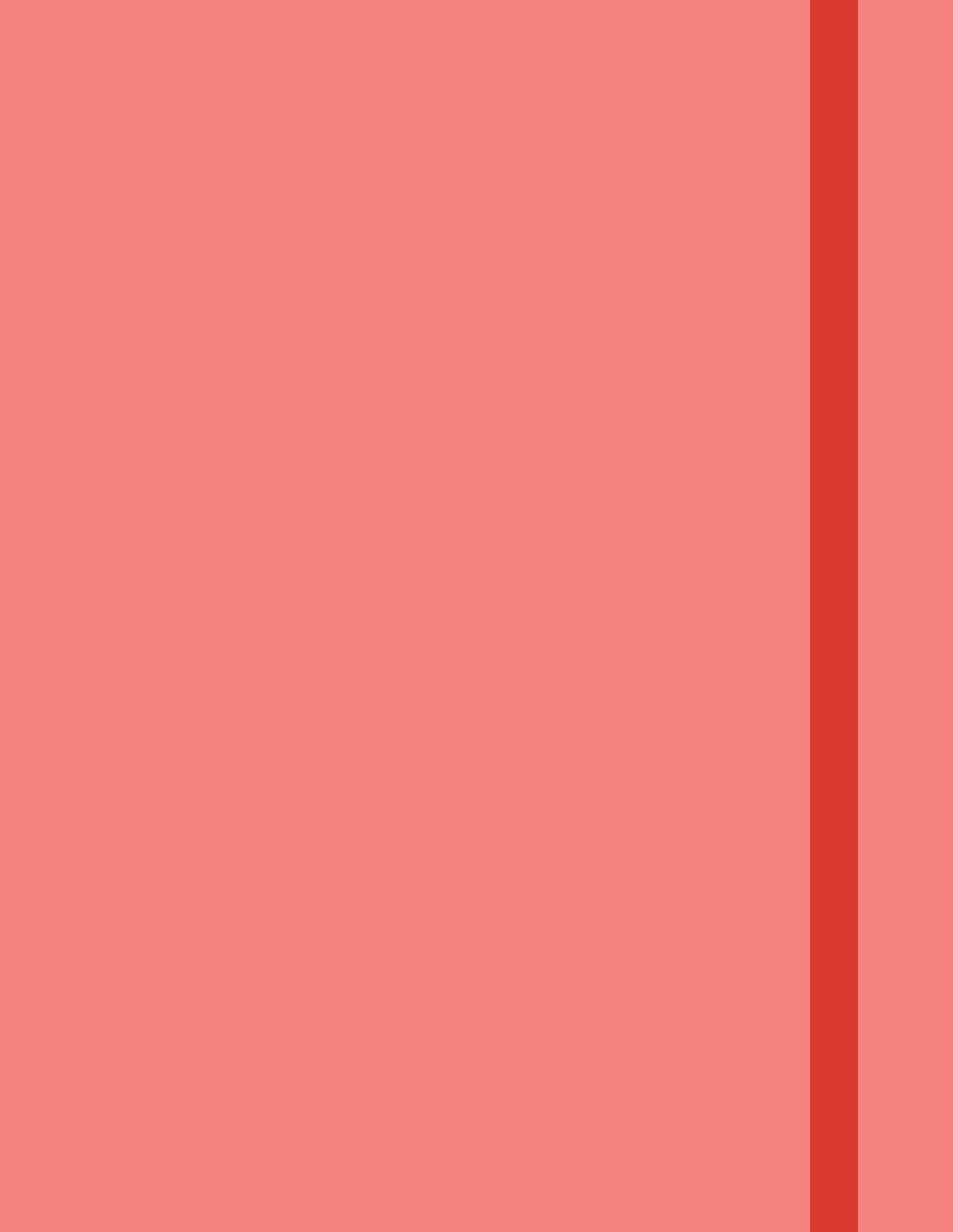
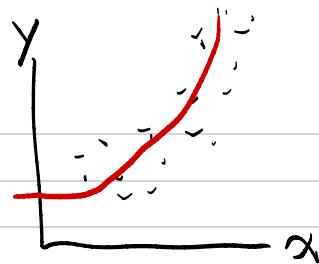




$$f(x) = c_0 + c_1 e^{c_2 x}$$

$$S = \sum (y - f(x))^2$$

$$= \sum (y - \underbrace{c_0 + c_1 e^{c_2 x}}_{f(x)})^2$$



$$c_0: \frac{\partial S}{\partial c_0} = 2 \sum (y - c_0 - c_1 e^{c_2 x}) (-1) \\ = -2 \sum (y - \underbrace{c_0 + c_1 e^{c_2 x}}_{f(x)}) \quad (1)$$

$$c_1: \frac{\partial S}{\partial c_1} = 2 \sum (y - c_0 - c_1 e^{c_2 x}) (-e^{c_2 x}) \\ = -2 \left(\sum_i e^{c_2 x_i} (y_i - \underbrace{c_0 + c_1 e^{c_2 x_i}}_{f(x_i)}) \right) \quad (2)$$

$$c_2: \frac{\partial S}{\partial c_2} = 2 \sum (y - c_0 - c_1 e^{c_2 x}) (-c_1 x e^{c_2 x}) \quad (3)$$

$$f(x) = c_0 + c_1 e^{c_2 x}$$

randomize c_0, c_1, c_2

$c_0^{(0)}, c_1^{(0)}, c_2^{(0)}$

calculate $f(x)$

calculate gradient
of current
iteration

✓
 $\frac{\partial S}{\partial c_0}, \frac{\partial S}{\partial c_1}, \frac{\partial S}{\partial c_2}$

grad- c_0 / grad- c_1
grad- c_2

α
step_size
learning
rate

Update

$c_0^{(t+1)} = c_0^{(t)} - \text{step_size} \times \text{grad-}c_0$

$c_1^{(t+1)} = c_1^{(t)} - \text{step_size} \times \text{grad-}c_1$

$c_2^{(t+1)} = c_2^{(t)} - \text{step_size} \times \text{grad-}c_2$

Stop?

End